

Inclusion Criteria and Thematic Scoring of Selected LLM-Agent Research Papers

Table 1: Inclusion Criteria for Paper Selection Based on Thematic Scoring

Ref.	Focus on LLM Agentic Behavior	Training Mechanisms for Autonomy	Interaction with External Tools or Environments	Frameworks for Single/Multi-Agent Systems	Reasoning, Planning, Memory, or Self-Reflection	Prompting, Fine-Tuning, or Memory Augmentation	Evaluation Metrics and Benchmarks	Challenges, Limitations, and Ethical Considerations	Total Score
[44]	8	8	8	8	8	8	8	6	62
[4]	6	8	8	8	7	7	6	6	56
[28]	8	8	8	8	8	8	7	8	63
[48]	8	8	8	8	8	8	8	5	61
[9]	8	8	8	7	7	5	6	8	57
[55]	5	5	8	6	8	8	7	8	55
[75]	8	8	8	8	8	8	7	8	63
[40]	6	7	7	8	7	6	8	6	55
[46]	8	7	8	5	5	6	7	8	54
[37]	8	8	8	8	8	8	8	6	62
[99]	6	7	8	8	8	5	5	7	54
[103]	8	8	6	5	6	8	7	5	53
[102]	8	8	8	8	8	8	8	7	63
[53]	6	5	7	8	8	6	8	7	55
[89]	8	8	8	8	8	8	8	5	61
[97]	7	7	6	8	7	8	5	6	54
[69]	6	7	7	5	7	8	8	8	56
[96]	8	8	8	8	8	8	7	8	63
[38]	6	6	6	7	5	8	8	8	54
[63]	8	8	7	6	6	8	5	7	55
[45]	8	8	8	8	8	8	8	6	62
[15]	6	5	7	7	6	8	7	7	53
[19]	8	8	8	8	8	8	8	7	63
[90]	5	8	6	7	6	8	8	8	56
[51]	8	6	7	5	6	7	7	8	54
[108]	8	8	8	8	8	8	8	5	61
[11]	5	5	6	8	7	8	8	8	55

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[93]	7	6	8	8	7	7	6	5	54
[60]	6	7	7	8	8	6	6	8	56
[33]	8	8	8	8	8	8	8	6	62
[70]	7	8	8	8	6	5	8	6	56
[67]	6	6	8	5	8	8	7	8	56
[81]	8	8	8	8	8	8	8	7	63
[30]	5	8	5	6	7	8	8	7	54
[73]	6	6	8	6	7	5	8	8	54
[13]	8	8	8	8	8	8	8	5	61
[68]	6	8	7	8	8	6	7	7	57
[26]	7	6	5	6	7	8	8	8	55
[88]	8	8	8	8	8	8	8	6	62
[83]	6	8	7	8	7	8	8	5	57
[39]	7	8	6	7	8	5	8	7	56
[59]	5	6	8	8	6	7	8	7	55
[91]	8	8	8	8	8	8	8	7	63
[25]	6	8	8	6	8	6	7	8	57
[2]	5	7	6	8	8	7	6	7	54
[41]	7	5	7	8	8	6	7	6	54
[58]	8	8	8	8	8	8	8	6	62
[43]	8	7	7	8	8	7	5	6	56
[95]	7	8	6	8	8	7	7	5	56
[66]	6	7	5	8	6	8	7	8	55
[42]	8	6	8	7	8	5	7	6	55
[52]	8	8	8	8	8	8	8	6	62
[34]	5	6	8	8	7	6	8	7	55
[7]	8	8	8	8	8	8	8	5	61
[57]	7	8	6	8	8	5	7	8	57
[3]	8	8	8	8	8	8	8	6	62
[106]	5	6	7	8	8	7	6	8	55
[76]	8	7	8	8	6	8	8	5	58
[105]	6	8	8	8	8	7	8	6	59
[27]	8	8	8	8	8	8	7	8	63
[12]	5	5	6	7	8	8	8	7	54
[10]	7	8	8	8	8	6	8	7	60
[78]	8	8	8	8	8	8	8	5	61

Ref.	Focus on LLM Agentic Behavior	Training Mechanisms for Autonomy	Interaction with External Tools or Environments	Frameworks for Single/Multi-Agent Systems	Reasoning, Planning, Memory, or Self-Reflection	Prompting, Fine-Tuning, or Memory Augmentation	Evaluation Metrics and Benchmarks	Challenges, Limitations, and Ethical Considerations	Total Score
[98]	6	7	8	5	8	7	8	8	57
[79]	8	8	7	8	8	8	8	6	61
[23]	5	8	8	6	7	8	8	8	58
[16]	8	8	8	8	8	8	8	7	63
[85]	7	6	5	8	8	8	7	8	57
[64]	8	8	8	8	8	8	6	8	62
[65]	6	8	7	8	5	8	7	8	57
[100]	8	8	8	8	8	8	8	5	61
[21]	5	7	8	8	6	8	8	7	57
[18]	8	8	8	8	8	8	7	8	63
[36]	8	6	8	7	8	8	8	5	58
[71]	8	8	8	8	8	8	8	6	62
[56]	7	8	5	8	8	6	8	8	58
[94]	8	8	8	8	8	8	8	7	63
[6]	6	5	8	8	7	8	8	8	58
[35]	8	8	8	8	8	8	6	8	62
[107]	8	7	8	5	8	8	8	8	60
[61]	8	8	8	8	8	8	8	5	61
[49]	5	8	6	8	8	7	8	8	58
[5]	8	8	8	8	8	8	7	8	63
[86]	7	6	8	8	8	5	8	8	58
[54]	8	8	8	8	8	8	8	6	62
[101]	8	5	7	8	8	8	6	8	58
[8]	8	8	8	8	8	8	8	7	63
[22]	6	8	8	7	8	8	5	8	58
[17]	8	8	8	8	8	8	6	8	62
[32]	5	8	8	8	6	8	8	7	58
[77]	8	8	8	8	8	8	8	5	61
[80]	7	8	8	5	8	8	8	8	60
[104]	8	8	8	8	8	8	7	8	63
[92]	8	6	8	8	8	5	8	8	59
[50]	8	8	8	8	8	8	8	6	62
[72]	5	7	8	8	8	8	6	8	58
[31]	5	8	5	6	7	8	8	7	54
[47]	6	6	8	6	7	5	8	8	54
[14]	8	8	8	8	8	8	8	5	61

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[29]	6	8	7	8	8	6	7	7	57
[74]	7	6	5	6	7	8	8	8	55
[82]	8	8	8	8	8	8	8	6	62
[84]	8	8	8	8	8	8	8	5	61
[87]	5	7	8	8	6	8	8	7	57
[20]	8	8	8	8	8	8	7	8	63
[24]	8	6	8	7	8	8	8	5	58
[1]	7	8	5	8	8	6	8	8	58
[62]	5	6	7	8	5	6	7	8	52

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